Exp: 5 **Packet Capture using Wireshark**

Aim:

To understand the features of Wireshark as a packet capture tool and to understand encapsulation of information at the various layers of the protocol stack.

Requirements:

* Computer with Wireshark installed
* Network connection

Procedure:

1. Open Wireshark and select the active interface (Wi-Fi / Ethernet).
2. Click Start to begin the capture; generate traffic (ping, browse a website).
3. Click Stop and apply display filters to study the packets:

arp	ARP packets
icmp	ping (ICMP) packets
dns	DNS queries and responses
http	HTTP traffic
tcp / udp	TCP / UDP packets

ip.addr == 192.168.1.1

tcp.port == 80

4. Select a packet and study the Packet Details pane: Frame (physical), Ethernet II (data link), Internet Protocol (network), TCP/UDP (transport) and the application layer data. This shows encapsulation at each layer.
5. Use Statistics > Flow Graph / Conversations to view the packet flow.
6. Save the capture using File > Save. Layers observed: Frame - Ethernet II (MAC addresses) - IPv4 (IP

addresses, TTL) - TCP/UDP (ports, flags) - Application (HTTP/DNS).

Result:

Packets were captured and analysed with Wireshark and encapsulation at the different layers was understood.